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Parameter–Elliptic Fourier Multipliers Systems and Generation of Analytic and C^∞ Semigroups

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Abstract: We consider Fourier multiplier systems on $\mathbb{R}^n$ with components belonging to the standard Hörmander class $S^m_{1,0}(\mathbb{R}^n)$, but with limited regularity. Using a notion of parameter-ellipticity with respect to a subsector $\Lambda \subset \mathbb{C}$ (introduced by Denk, Saal, and Seiler) we show the generation of both C^∞ semigroups and analytic semigroups (in a particular case) on the Sobolev spaces $W^k_p(\mathbb{R}^n, \mathbb{C}^q)$ with $k \in \mathbb{N}_0$, $1 \leq p < \infty$ and $q \in \mathbb{N}$. For the proofs, we modify and improve a crucial estimate from Denk, Saal and Seiler, on the inverse matrix of the symbol (see Lemma 2). As examples, we apply the theory to solve the heat equation, a linear thermoelastic plate equation, a structurally damped plate equation, and a generalized plate equation, all in the whole space, in the frame of Sobolev spaces.

Keywords: C^∞-semigroups; analytic semigroups; Fourier multipliers; Λ-ellipticity

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1. Introduction

Elliptic systems of partial differential equations were introduced in 1955 by A. Douglis and L. Nirenberg in [1]. Then, in 1973, R. Kramer formulated and solved in [2] several Cauchy problems for systems of partial differential equations which are elliptic in the sense given by Douglis and Nirenberg in [1]. In the same year, A. Koževnikov, in his study in [3] about spectral asymptotics for elliptic pseudodifferential systems with the structure of Douglis–Nirenberg, introduced an algebraic condition on the symbol (called the parameter–ellipticity condition) which permitted him to prove the similarity of the system satisfying this condition to an almost diagonal system up to a symbol of order $-\infty$, but he did not consider questions of equation solvability for those operators. In 2009, R. Denk, J. Saal and J. Seiler considered in [4] pseudodifferential Douglis–Nirenberg systems on $\mathbb{R}^n$ with components belonging to the standard Hörmander class $S^*_{1,\delta}(\mathbb{R}^n \times \mathbb{R}^n)$, $0 \leq \delta < 1$. They introduced the formulation of parameter–ellipticity with respect to a subsector $\Lambda \subset \mathbb{C}$, which is motivated by a notion of parameter–ellipticity introduced by Denk, Menniken, and Volevich in [5] and connected with the so-called Newton polygon associated with the system. They showed that their formulation of ellipticity is equivalent to the given by Koževnikov in [3] and that this condition implies the existence of a bounded H_∞-calculus for their pseudodifferential systems in suitable scales of Sobolev spaces with $1 < p < \infty$, hence of L_p-maximal regularity. Furthermore, it is known that the maximal regularity implies the generation of an analytic semigroup, however the reverse implication is false.

In this paper, we will consider certain Fourier multiplier systems on $\mathbb{R}^n$, similar but not necessarily with the exact structure of a Douglis–Nirenberg system, with components belonging to the standard Hörmander class $S^m_{1,0}(\mathbb{R}^n)$, but with limited regularity (see Definition 2), and using the notion of parameter–ellipticity with respect to a subsector $\Lambda \subset \mathbb{C}$ given in [4], we will establish (in Theorem 1) the generation of C^∞ semigroups and

analytic semigroups (in a particular case) on the Sobolev spaces $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ with $k \in \mathbb{N}_0$ and $1 \leq p < \infty$ giving a direct proof. For this direct proof of our main result we use the approach based on oscillatory integrals and kernel estimates for them (as in [6]), taking advantage of the fact that the associated symbols to the pseudodifferential operators are matrices valued and the entries of these matrices are symbols of order greater than $1/2$ and are independent of the spatial variable. An application to non-autonomous pseudodifferential Cauchy problems gives the existence and uniqueness of a classical solution (see Theorem 2). As examples, we apply the theory to solve the heat equation, a linear thermoelastic plate equation, a structurally damped plate equation, and a generalized plate equation, all in the whole space, in suitable Sobolev spaces (see Section 5). Other applications of the theory of semigroups and its generalizations address the control and stablility theory for mechanical systems or the controllability of fractional evolution equations or inclusions (see [7–14] and the conclusions in Section 6).

The paper is organized as follows: In Section 2 we present the definition of our system of Fourier multipliers, which are defined in terms of suitable oscillatory integrals. Following [4], we give in Section 3 the notion of Λ ellipticity for this system of Fourier multipliers, with respect to a sector Λ of the complex plane. In order to allow that the correspondent estimate in the definition of Λ ellipticity for the characteristic polynomial of the matrix symbol of our system of Fourier multipliers hold for all values of the symbol variable ξ in $\mathbb{R}^n$, we consider a perturbation of the system by a constant, following again the ideas given in [4] (see Remark 2). Section 4 is the core of the paper. There we obtain the main result of the paper about generation, under suitable hypothesis, of C^∞ semigroups and analytic semigroups for a Sobolev space realization of the perturbed operator associated to a Λ-elliptic system (Theorem 1). We also present in that section, existence and uniqueness results for non-autonomous Cauchy problems based on the obtained results about generation of semigroups (Theorem 2 and corollary 2). In Section 5, as examples and as already mentioned above, the heat equation, a linear thermoelastic plate equation, a structurally damped plate equation, and a generalized plate equation are considered. Finally, in the conclusions in Section 6, we summarize the results obtained in the paper and point out some possible future scope of this work.

2. Fourier Multiplier Systems

In the following, for $n \in \mathbb{N}$, ρ_n denotes the smallest even integer greater than n, E represents an arbitary Banach space, $\mathcal{L}(E)$ the space of linear and continuous maps of E into E, $\mathcal{S}(\mathbb{R}^n, E)$ the Schwartz space of rapidly decreasing functions and $C_b^\infty(\mathbb{R}^n, E)$ the space of all functions $u : \mathbb{R}^n \to E$ such that $\partial^\alpha u$ is bounded and continuous on $\mathbb{R}^n$ for all $\alpha \in \mathbb{N}_0^n$. $W_p^k(\mathbb{R}^n, E)$, for $k \in \mathbb{N}_0$ and $1 \leq p \leq \infty$, are the usual Sobolev spaces equipped with their standard norm and it is well konwn that $\mathcal{S}(\mathbb{R}^n, E) \subset C_b^\infty(\mathbb{R}^n, E) \cap W_p^k(\mathbb{R}^n, E)$ and that $\mathcal{S}(\mathbb{R}^n, E)$ is dense in $W_p^k(\mathbb{R}^n, E)$ if $1 \leq p < \infty$. Also we will use the following notations throughout the paper: $D_{\xi_j} := -i\partial_{\xi_j}$, $\langle \xi \rangle := (1 + |\xi|^2)^{1/2}$, $\langle \xi, t \rangle := (1 + |\xi|^2 + |t|^2)^{1/2}$ and $|\xi, t| := (|\xi|^2 + |t|^2)^{1/2}$, for $\xi \in \mathbb{R}^n$ and $t \in \mathbb{R}$.

For the following definition, see Equation (1) in [6].

Definition 1. *Let $m \in \mathbb{R}$ and $\rho \in \mathbb{N}_0$.*

(a) *The symbol class $S^{m,\rho}(\mathbb{R}^n, \mathcal{L}(E)) := S_{1,0}^{m,\rho}(\mathbb{R}^n, \mathcal{L}(E))$ consists of all functions $a : \mathbb{R}^n \to \mathcal{L}(E)$ of class C^ρ with the property that for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq \rho$, there exists a positive constant C_α such that*

$$\left\| \partial_\xi^\alpha a(\xi) \right\|_{\mathcal{L}(E)} \leq C_\alpha \langle \xi \rangle^{m-|\alpha|} \text{ for all } \xi \in \mathbb{R}^n.$$

(b) *In $S^{m,\rho}(\mathbb{R}^n, \mathcal{L}(E))$ we define the norm*

$$\|a\|_{S^{m,\rho}} := \max_{|\alpha| \leq \rho} \sup_{\xi \in \mathbb{R}^n} \langle \xi \rangle^{|\alpha|-m} \left\| \partial_\xi^\alpha a(\xi) \right\|_{\mathcal{L}(E)}.$$

(c) *For $a \in S^{m,\rho}(\mathbb{R}^n, \mathcal{L}(E))$ with $\rho \geq \rho_n$, the Fourier multiplier operator $a(D)$ is defined by*

$$[a(D)u](x) := \mathrm{Os}-\iint e^{i\xi\cdot\eta} a(\xi) u(x-\eta) \frac{d(\xi,\eta)}{(2\pi)^n} \tag{1}$$

for all $x \in \mathbb{R}^n$ and $u \in C_b^\infty(\mathbb{R}^n, E)$, where the symbol $\mathrm{Os}-\iint$ stands for oscillatory integrals.

In the case that $E = \mathbb{C}^q$, $q \in \mathbb{N}$, we identify $\mathcal{L}(\mathbb{C}^q)$ with $\mathbb{C}^{q\times q}$, $\mathbb{C}^{1\times 1}$ with $\mathbb{C}$ and we write $S^{m,\rho}(\mathbb{R}^n)$ instead of $S^{m,\rho}(\mathbb{R}^n, \mathbb{C})$.

Remark 1. (a) *The definition and some properties of the oscillatory integrals can be found in [15] for the scalar case and in [16] (Appendix A) for the vector valued case.*

(b) *For $\rho \geq \rho_n$, Lemma A.4 and Remark A.5 in [16] imply that the oscillatory integral in (1) exists. Moreover, due to Lemma A.6 in [16] we have that $a(D) \in \mathcal{L}(C_b^\infty(\mathbb{R}^n, E))$.*

(c) *Fourier multipliers with limited regularity symbols were also studied in [17,18].*

Definition 2 (Compare with [4] (Definition 2.3))**.** *The Fourier multipliers system we will consider in this paper is a $q \times q$-matrix of Fourier multipliers*

$$A(D) = \big(a_{ij}(D)\big)_{1\leq i,j\leq q}$$

such that

$$a_{ij} \in S^{r_{ij},\rho}(\mathbb{R}^n),$$

where $r_{ij} \in \mathbb{R}$, $r_i := r_{ii} \geq 0$, for all $i, j = 1, ..., q$, and $\rho \in \mathbb{N}$ is such that $\rho \geq \rho_n$.

3. Λ-Elliptic Fourier Multipliers Systems

From now on we fix θ, with $0 < \theta < \pi$, and let $\Lambda(\theta)$ denote the closed subsector of the complex plane $\mathbb{C}$, given by

$$\Lambda := \Lambda(\theta) := \Big\{ re^{i\gamma} : r \geq 0, \theta \leq \gamma \leq 2\pi - \theta \Big\}.$$

For the following definition we refer to [4] (Definition 3.1).

Definition 3. *Let $A(D)$ be a Fourier multipliers system (as in Definition 2). We say that $A(D)$ is* Λ-elliptic *(or* $\Lambda(\theta)$-elliptic *to highlight the angle) , if there exist constants $C > 0$ and $R \geq 0$ such that*

$$|p(\xi;\lambda)| \geq C\big(\langle\xi\rangle^{r_1} + |\lambda|\big) \cdots \big(\langle\xi\rangle^{r_q} + |\lambda|\big)$$

for all $(\xi, \lambda) \in \mathbb{R}^n \times \Lambda$ with $|\xi| \geq R$, where $p(\xi;\lambda) := \det(A(\xi) - \lambda)$.

Remark 2. *Let $A(D)$ be a Λ-elliptic Fourier multipliers system. Due to Lemma 3.4 in [4], there exists a constant $\alpha_0 \geq 0$ such that*

$$|\det(A_{\alpha_0}(\xi) - \lambda)| \geq C(\langle\xi\rangle^{r_1} + |\lambda|) \cdots (\langle\xi\rangle^{r_q} + |\lambda|) \quad \forall \xi \in \mathbb{R}^n \text{ and } \lambda \in \Lambda,$$

where $A_{\alpha_0}(\xi) := A(\xi) + \alpha_0$, i.e., $A_{\alpha_0}(D)$ is Λ-elliptic with $R = 0$.

Lemma 1 ([4], Lemma 3.5)**.** *Let $A(D)$ be Λ-elliptic and*

$$\big(g_{ij}(\xi;\lambda)\big)_{1\leq i,j\leq q} := (A_{\alpha_0}(\xi) - \lambda)^{-1}.$$

Then,

$$\left|\partial_\xi^\alpha g_{ij}(\xi;\lambda)\right| \leq C_\alpha \left(\langle\xi\rangle^{r_i} + |\lambda|\right)^{-1} \left(\langle\xi\rangle^{r_j} + |\lambda|\right)^{-1} \langle\xi\rangle^{r_{ij}-|\alpha|}, \quad (i \neq j)$$
$$\left|\partial_\xi^\alpha g_{ii}(\xi;\lambda)\right| \leq C_\alpha \left(\langle\xi\rangle^{r_i} + |\lambda|\right)^{-1} \langle\xi\rangle^{-|\alpha|}$$

for all $\alpha \in \mathbb{N}_0^n$*, being the estimates uniform in* $(\xi,\lambda) \in \mathbb{R}^n \times \Lambda$.

Following the ideas of the proof of this lemma in [4], we note that the condition $r_1 \geq \cdots \geq r_q \geq 0$ given there, is not necessary for the estimates above. However, we get another crucial estimate under the following additional assumption about the orders of the symbols in the system:

$$\sum_{j=1}^{k} r_{i_j \pi(i_j)} = \sum_{j=1}^{k} r_{i_j} \tag{2}$$

for all subsets of indices $\{i_1,\ldots,i_k\} \subset \{1,\ldots,q\}$ and all bijections $\pi : \{i_1,\ldots,i_k\} \to \{i_1,\ldots,i_k\}$.

The crucial estimate we mentioned above is given in the following lemma.

Lemma 2. *Let* $A(D)$ *be* Λ*-elliptic,*

$$\left(g_{ij}(\xi;\lambda)\right)_{1\leq i,j\leq q} := \left(A_{\alpha_0}(\xi) - \lambda\right)^{-1},$$

and suppose that the assumption (2) *holds. Then, for all* $i = 1,\ldots,q$, $\alpha \in \mathbb{N}_0^n$ *with* $0 < |\alpha| \leq \rho$, *and* $(\xi,\lambda) \in \mathbb{R}^n \times \Lambda$, *we have*

$$\left|\partial_\xi^\alpha g_{ii}(\xi;\lambda)\right| \leq C_\alpha \sum_{j=1}^{q} \left(\langle\xi\rangle^{r_i} + |\lambda|\right)^{-1} \left(\langle\xi\rangle^{r_j} + |\lambda|\right)^{-1} \langle\xi\rangle^{r_j-|\alpha|} \tag{3}$$

for some constant C_α.

Proof. Let $i \in \{1,\ldots,q\}$ be fixed. It should first be noted that

$$g_{ii}(\xi;\lambda) = \frac{1}{\det(A_{\alpha_0}(\xi) - \lambda)} \mathrm{Cof}_{(i,i)}\left(A_{\alpha_0}(\xi) - \lambda\right),$$

where $\mathrm{Cof}_{(i,i)}\left(A_{\alpha_0}(\xi) - \lambda\right)$ is the *cofactor* (i,i) of $A_{\alpha_0}(\xi) - \lambda$, that is, the determinant of the matrix obtained by removing the i-th row and i-th column of this matrix. With the convention $\prod_{l=k}^{m}(\cdots)_l := 1$ if $k > m$, which we will use from now on in this proof, we have that $\mathrm{Cof}_{(i,i)}\left(A_{\alpha_0}(\xi) - \lambda\right)$ is a linear combination of terms

$$\Big(\prod_{l=1}^{k}(a_{i_l i_l} + \alpha_0 - \lambda)\Big) \prod_{l=k+1}^{q-1} a_{i_l \pi(i_l)},$$

where $\{i_1,\ldots,i_{q-1}\} = \{1,\ldots,q\} \setminus \{i\}$, $0 \leq k \leq q-1$, and $\pi : \{i_1,\ldots,i_{q-1}\} \to \{i_1,\ldots,i_{q-1}\}$ is a bijection which have $\{i_1,\ldots,i_k\}$ as its set of fixed points. Therefore $\{i_{k+1},\ldots,i_{q-1}\} = \{\pi(i_{k+1}),\ldots,\pi(i_{q-1})\}$ and, in virtue of assumption (2), it holds

$$r_{i_{k+1}\pi(i_{k+1})} + \cdots + r_{i_{q-1}\pi(i_{q-1})} = r_{i_{k+1}} + \cdots + r_{i_{q-1}}. \tag{4}$$

If $\alpha \in \mathbb{N}_0^n$ with $0 < |\alpha| \leq \rho$, the Leibniz' formula implies that $\partial_\xi^\alpha g_{ii}$ is a linear combination of terms

$$\partial^\beta\Big(\frac{1}{p_0}\Big)\underbrace{\Big(\prod_{l=1}^{k}\partial^{\gamma_l}(a_{i_li_l}+\alpha_0-\lambda)\Big)\prod_{l=k+1}^{q-1}\partial^{\gamma_l}a_{i_l\pi(i_l)}}_{=:H}, \tag{5}$$

where $\beta, \gamma_1, \ldots, \gamma_{q-1} \in \mathbb{N}_0^n$ with $\beta + \gamma_1 + \cdots + \gamma_{q-1} = \alpha$, $k \in \{0, 1, \ldots, q\}$, and $p_0 = p_0(\xi, \lambda) := \det(A_{\alpha_0}(\xi) - \lambda)$. Note that the term $a_{ii} + \alpha_0 - \lambda$ is not in H (see (5)), and also we can estimate $\big|\partial^{\gamma_l}\big(a_{i_li_l} + \alpha_0 - \lambda\big)\big|$ from above by $\langle\xi\rangle^{r_{i_l}} + |\lambda|$ if $\gamma_l = 0$ and by $\langle\xi\rangle^{r_{i_l}-|\gamma_l|}$ if $\gamma_l \neq 0$.

If $\beta = 0$, then $\gamma_j \neq 0$ for some $j = 1, \ldots, q-1$. Therefore, the term related to $a_{i_ji_j}$ which appears in H is equal to $\partial^{\gamma_j}a_{i_ji_j}$, and then, due to the Λ-ellipticity condition (together with Remark 2) and (4), the expression (5) can be estimated from above by $\big(\langle\xi\rangle^{r_i} + |\lambda|\big)^{-1}\Big(\langle\xi\rangle^{r_{i_j}} + |\lambda|\Big)^{-1}\langle\xi\rangle^{r_{i_j}-|\alpha|}$.

In order to consider the case $\beta \neq 0$, we will prove first that for each $\alpha \in \mathbb{N}_0^n$, $0 < |\alpha| \leq \rho$, there exists $C > 0$ such that

$$|\partial^\alpha p_0(\xi,\lambda)| \leq C\Big(\sum_{j=1}^{q}\Big(\prod_{\substack{i=1\\ i\neq j}}^{q}(\langle\xi\rangle^{r_i} + |\lambda|)\Big)\langle\xi\rangle^{r_j}\Big)\langle\xi\rangle^{-|\alpha|} \tag{6}$$

for all $\xi \in \mathbb{R}^n$ and $\lambda \in \Lambda$. Let $\mathcal{Z} := \{1, \ldots, q\}$. Note that p_0 is a linear combination of terms of the form

$$\Big(\prod_{l=1}^{k}(a_{i_li_l} + \alpha_0 - \lambda)\Big)\prod_{l=k+1}^{q}a_{i_l\pi(i_l)} \qquad (k = 0, \ldots, q),$$

where $\pi : \mathcal{Z} \to \mathcal{Z}$ is a bijection with fixed points $i_1, \ldots, i_k$, and therefore $\{i_{k+1}, \ldots, i_q\} = \{\pi(i_{k+1}), \ldots, \pi(i_q)\}$ which, again due to the assumption (2), yields

$$r_{i_{k+1}\pi(i_{k+1})} + \cdots + r_{i_q\pi(i_q)} = r_{k+1} + \cdots + r_q. \tag{7}$$

Indeed, if $\mathcal{P}_k$, $k = 0, 1, \ldots, q$, denotes the set of all bijections $\pi : \mathcal{Z} \to \mathcal{Z}$ with exactly k fixed points, then p_0 can be written as

$$p_0 = \sum_{k=0}^{q-1}\sum_{\substack{\pi\in\mathcal{P}_k\\ i_1<\cdots<i_k\\ i_{k+1}<\cdots<i_q}} \pm\Big(\prod_{l=1}^{k}(a_{i_li_l} + \alpha_0 - \lambda)\Big)\prod_{l=k+1}^{q}a_{i_l\pi(i_l)},$$

where in each summand, $i_1, \ldots, i_k$ are the fixed points of π.

If $\alpha \in \mathbb{N}_0^n$, $0 < |\alpha| \leq \rho$, then

$$\partial^\alpha p_0 = \sum_{k=0}^{q-1}\sum_{\substack{\pi\in\mathcal{P}_k\\ i_1<\cdots<i_k\\ i_{k+1}<\cdots<i_q}}\sum_{\substack{\alpha_1,\ldots,\alpha_q\in\mathbb{N}_0^n\\ \alpha_1+\cdots\alpha_q=\alpha}} C_{\alpha_1\ldots\alpha_q}\underbrace{\Big(\prod_{l=1}^{k}\partial^{\alpha_l}(a_{i_li_l} + \alpha_0 - \lambda)\Big)\prod_{l=k+1}^{q}\partial^{\alpha_l}a_{i_l\pi(i_l)}}_{=:Q_k}.$$

Now,

$$\big|\partial^{\alpha_l}(a_{i_li_l} + \alpha_0 - \lambda)\big| \leq C\begin{cases}\langle\xi\rangle^{r_{i_l}} + |\lambda|, & \alpha_l = 0,\\ \langle\xi\rangle^{r_{i_l}}\langle\xi\rangle^{-|\alpha_l|}, & \alpha_l \neq 0,\end{cases}$$

and

$$\big|\partial^{\alpha_l}a_{i_l\pi(i_l)}\big| \leq C\langle\xi\rangle^{r_{i_l\pi(i_l)}}\langle\xi\rangle^{-|\alpha_l|}.$$

Since $\alpha \neq 0$, $\alpha_j \neq 0$ for some j and, therefore, taking (7) in account, it holds

$$|Q_k| \leq C\Big(\prod_{\substack{l=1\\ l\neq j}}^{q}\big(\langle\xi\rangle^{r_{i_l}} + |\lambda|\big)\Big)\langle\xi\rangle^{r_{i_j}-|\alpha|}.$$

Then, we have

$$\begin{aligned}|\partial^\alpha p_0| &\leq \overline{C}\sum_{k=0}^{q-1}\sum_{\pi\in\mathcal{P}_k}\sum_{j=1}^{q}\Big(\prod_{\substack{l=1\\ l\neq j}}^{q}\big(\langle\xi\rangle^{r_{i_l}} + |\lambda|\big)\Big)\langle\xi\rangle^{r_{i_j}}\langle\xi\rangle^{-|\alpha|}\\ &\leq \hat{C}\Big(\sum_{j=1}^{q}\Big(\prod_{\substack{i=1\\ i\neq j}}^{q}\big(\langle\xi\rangle^{r_i} + |\lambda|\big)\Big)\langle\xi\rangle^{r_j}\Big)\langle\xi\rangle^{-|\alpha|},\end{aligned}$$

which shows (6). Thus, we can estimate $\partial^\beta(1/p_0)$ for $\beta \neq 0$. Indeed, if $\beta \neq 0$ it holds that (see [19], Lemma 10.4, p. 74)

$$\partial^\beta\left(\frac{1}{p_0}\right) = \sum_{k=1}^{|\beta|}\sum_{\substack{\beta_1,\dots,\beta_k\in\mathbb{N}_0^n\setminus\{0\}\\ \beta_1+\cdots+\beta_k=\beta}} C_{\beta_1\dots\beta_k}\frac{(\partial^{\beta_1}p_0)\cdots(\partial^{\beta_k}p_0)}{p_0^{1+k}}.$$

Due to (6) and the Λ-ellipticity condition we obtain

$$\begin{aligned}\left|\partial^\beta\left(\frac{1}{p_0}\right)\right| &\leq C_\beta\sum_{k=1}^{|\beta|}\sum_{\substack{\beta_1,\dots,\beta_k\in\mathbb{N}_0^n\setminus\{0\}\\ \beta_1+\cdots+\beta_k=\beta}}\frac{\Big(\sum_{j=1}^{q}\Big(\prod_{\substack{i=1\\ i\neq j}}^{q}\big(\langle\xi\rangle^{r_i}+|\lambda|\big)\Big)\langle\xi\rangle^{r_j}\Big)^k\langle\xi\rangle^{-|\beta|}}{|p_0|^{1+k}}\\ &\leq \hat{C}_\beta\sum_{k=1}^{|\beta|}\frac{\Big(\sum_{j=1}^{q}\Big(\prod_{\substack{i=1\\ i\neq j}}^{q}\big(\langle\xi\rangle^{r_i}+|\lambda|\big)^k\Big)\langle\xi\rangle^{kr_j}\Big)\langle\xi\rangle^{-|\beta|}}{\prod_{i=1}^{q}\big(\langle\xi\rangle^{r_i}+|\lambda|\big)^{1+k}}\\ &= \frac{\hat{C}_\beta}{\prod_{i=1}^{q}\big(\langle\xi\rangle^{r_i}+|\lambda|\big)}\sum_{k=1}^{|\beta|}\sum_{j=1}^{q}\frac{\langle\xi\rangle^{kr_j}\langle\xi\rangle^{-|\beta|}}{\big(\langle\xi\rangle^{r_j}+|\lambda|\big)^k}\\ &= \frac{\hat{C}_\beta}{\prod_{i=1}^{q}\big(\langle\xi\rangle^{r_i}+|\lambda|\big)}\sum_{k=1}^{|\beta|}\sum_{j=1}^{q}\Big(\frac{\langle\xi\rangle^{r_j}}{\langle\xi\rangle^{r_j}+|\lambda|}\Big)^{k-1}\frac{\langle\xi\rangle^{r_j}\langle\xi\rangle^{-|\beta|}}{\langle\xi\rangle^{r_j}+|\lambda|}\\ &\leq \frac{C^*_\beta}{\prod_{i=1}^{q}\big(\langle\xi\rangle^{r_i}+|\lambda|\big)}\Big(\sum_{j=1}^{q}\frac{\langle\xi\rangle^{r_j}}{\langle\xi\rangle^{r_j}+|\lambda|}\Big)\langle\xi\rangle^{-|\beta|}.\end{aligned}$$

Now, since

$$|H| \leq \Big(\prod_{\substack{l=1\\ l\neq i}}^{q}\big(\langle\xi\rangle^{r_l}+|\lambda|\big)\Big)\langle\xi\rangle^{-|\gamma_1+\cdots+\gamma_{q-1}|},$$

if $0 < |\beta| \leq \rho$, we can estimate (5) from above by

$$\frac{C^*_\beta}{\prod\limits_{i=1}^{q}\left(\langle\xi\rangle^{r_i}+|\lambda|\right)}\Big(\sum_{j=1}^{q}\frac{\langle\xi\rangle^{r_j}}{\langle\xi\rangle^{r_j}+|\lambda|}\Big)\Big(\prod_{\substack{l=1\\ l\neq i}}^{q}\left(\langle\xi\rangle^{r_l}+|\lambda|\right)\Big)\langle\xi\rangle^{-|\alpha|}$$
$$= C^*_\beta\sum_{j=1}^{q}\frac{\langle\xi\rangle^{r_j-|\alpha|}}{\left(\langle\xi\rangle^{r_i}+|\lambda|\right)\left(\langle\xi\rangle^{r_j}+|\lambda|\right)}.$$

With the estimates from above for (5), in both cases $\beta = 0$ and $\beta \neq 0$, we obtain the estimate (3) for $0 < |\alpha| \leq \rho$ and $(\xi,\lambda) \in \mathbb{R}^n \times \Lambda$. □

Under the assumption (2) on the order of the symbols in the system, Lemma 1, estimate (3), and the equivalence

$$\langle\xi\rangle^r + |\lambda| \sim \left\langle \xi, |\lambda|^{1/r}\right\rangle^r \qquad (r \geq 0),$$

lead to the following assertion.

Corollary 1. *Let $A(D)$ be Λ-elliptic,*

$$b_\lambda(\cdot) := \left(A_{\alpha_0}(\cdot) - \lambda\right)^{-1} = \left(g_{ij}(\cdot;\lambda)\right)_{1\leq i,j\leq q},$$

and suppose that the assumption (2) holds. Then for each $i, j = 1, \ldots, q$, we have

$$\left(b_\lambda(\cdot)\right)_{ij} = g_{ij}(\cdot;\lambda) \in S^{-r_{ji},\rho}(\mathbb{R}^n, \mathcal{L}(\mathbb{C}^q)), \quad \forall\lambda \in \Lambda$$

with

$$|g_{ii}(\xi;\lambda)| \leq C\left\langle \xi, |\lambda|^{1/r_i}\right\rangle^{-r_i}, \qquad (i = 1, ..., q)$$
$$\left|\partial^\alpha_\xi g_{ii}(\xi;\lambda)\right| \leq C_\alpha\sum_{j=1}^{q}\left\langle \xi, |\lambda|^{1/r_i}\right\rangle^{-r_i}\left\langle \xi, |\lambda|^{1/r_j}\right\rangle^{-r_j}\langle\xi\rangle^{r_j-|\alpha|}, \quad ((i,\alpha) \in \mathcal{Z}_1)$$
$$\left|\partial^\alpha_\xi g_{ij}(\xi;\lambda)\right| \leq C\left\langle \xi, |\lambda|^{1/r_i}\right\rangle^{-r_i}\left\langle \xi, |\lambda|^{1/r_j}\right\rangle^{-r_j}\langle\xi\rangle^{r_{ij}-|\alpha|}, \quad ((i,j,\alpha) \in \mathcal{Z}_2)$$

for all $(\xi,\lambda) \in \mathbb{R}^n \times \Lambda$, where C is a positive constant independent on α, ξ and λ, $\mathcal{Z}_1 := \{(i,\alpha) : 1 \leq i \leq q,\ 0 < |\alpha| \leq \rho\}$, and $\mathcal{Z}_2 := \{(i,j,\alpha) : 1 \leq i,j \leq q,\ i \neq j,\ |\alpha| \leq \rho\}$.

4. Generation of Analytic and C^∞-Semigroups

In this section, under the assumption (2) on the order of the symbols, we will prove the main result of this paper (Theorem 1). For that we will need to estimate the norm $\|b_\lambda(D)u\|_{W^k_p(\mathbb{R}^n,\mathbb{C}^q)}$.

Let $A(D)$ be Λ-elliptic with $r_{ij} > 0$, $\rho \geq \rho_n$, and suppose that the assumption (2) holds. Then, note that $r_{ij} + r_{ji} = r_i + r_j$ for $i, j = 1, \ldots, q$. Moreover, let $r^+ := \max\limits_{1\leq i\leq q}\{r_i\}$, $r_- := \min\limits_{1\leq i,j\leq q}\{r_{ij}\}$, $\omega \geq 1$ and

$$\Lambda_\omega := \Lambda(\theta)_\omega := \{\lambda \in \Lambda = \Lambda(\theta) : |\lambda| \geq \omega\}.$$

Note that for b_λ, as in corollary 1, $u \in C^\infty_b(\mathbb{R}^n,\mathbb{C}^q) \cap W^k_p(\mathbb{R}^n,\mathbb{C}^q)$ and $\beta \in \mathbb{N}^n_0$, we have

$$\partial_x^\beta (b_\lambda(D)u)(x) = \mathrm{Os}\!-\!\iint e^{i\xi\cdot\eta} b_\lambda(\xi)(\partial_x^\beta u)(x-\eta)\frac{d(\xi,\eta)}{(2\pi)^n}$$
$$= \lim_{\varepsilon\searrow 0}\int_{\mathbb{R}^n} K_\varepsilon(\eta,\lambda)(\partial_x^\beta u)(x-\eta)d\eta \tag{8}$$

with

$$K_\varepsilon(\eta,\lambda) := \int_{\mathbb{R}^n} e^{i\xi\cdot\eta}\chi_\varepsilon(\xi,\eta;\lambda)b_\lambda(\xi)\frac{d\xi}{(2\pi)^n}, \tag{9}$$

and

$$\chi_\varepsilon(\xi,\eta;\lambda) := \chi_\varepsilon(\xi;\lambda)\psi_\varepsilon(\eta)$$

for $\xi,\eta\in\mathbb{R}^n$, $0<\varepsilon<1$, where ψ is a function in $\mathcal{S}(\mathbb{R}^n)$ with $\psi(0)=1$, $\psi_\varepsilon(\eta):=\psi(\varepsilon\eta)$, $\chi_\varepsilon(\xi;\lambda) := \varphi_\varepsilon\big((|\xi|^2+|\lambda|^{2/r^+})^{1/2}\big)$ with $\varphi_\varepsilon(x):=\varphi(\varepsilon x)$ for $x\in\mathbb{R}$ and $\varphi\in\mathcal{S}(\mathbb{R})$ satisfies $\varphi(0)=1$.

It was proven in [20] (p. 845) that for $\alpha\in\mathbb{N}_0^n$, there exists a constant $C_\alpha>0$ such that for all $\xi\in\mathbb{R}^n$ and $\lambda\in\Lambda_\omega$,

$$\left|\partial_\xi^\alpha\chi_\varepsilon(\xi;\lambda)\right| \le C_\alpha\big(|\xi|^2+|\lambda|^{2/r^+}\big)^{-|\alpha|/2} \qquad (0<\varepsilon<1).$$

Now, due to

$$\frac{\omega^{2/r^+}+1}{\omega^{2/r^+}}\big(|\xi|^2+|\lambda|^{2/r^+}\big) \ge \frac{1}{\omega^{2/r^+}}\big(\omega^{2/r^+}|\xi|^2+\omega^{2/r^+}|\lambda|^{2/r^+}+|\lambda|^{2/r^+}\big)$$
$$\ge |\xi|^2+|\lambda|^{2/r^+}+1,$$

we have

$$\left|\partial_\xi^\alpha\chi_\varepsilon(\xi;\lambda)\right| \le \overline{C}_\alpha\left\langle\xi,|\lambda|^{1/r^+}\right\rangle^{-|\alpha|} \qquad (0<\varepsilon<1). \tag{10}$$

We will obtain some estimate for K_ε with help of (10) and the following lemma and remark.

Lemma 3 ([15], Lemma 6.3)**.** *Let $\chi\in\mathcal{S}(\mathbb{R}^n)$ with $\chi(0)=1$. Then:*

(a) $\chi(\varepsilon x)\underset{\varepsilon\searrow 0}{\longrightarrow} 1$ *uniformly on all compact subset of* $\mathbb{R}^n$.

(b) $\partial_x^\alpha\chi(\varepsilon x)\underset{\varepsilon\searrow 0}{\longrightarrow} 0$ *uniformly on* $\mathbb{R}^n$, *if* $\alpha\neq 0$.

(c) *For all $\alpha\in\mathbb{N}_0^n$, there exists some $C_\alpha>0$, independent on $0<\varepsilon<1$, such that*

$$|\partial_x^\alpha\chi(\varepsilon x)| \le C_\alpha\langle x\rangle^{-(|\alpha|-\sigma)} \text{ for all } x\in\mathbb{R}^n \text{ and } 0\le\sigma\le|\alpha|.$$

Remark 3. *Note that, if $\frac{1}{2}<r_-$, then for all $\frac{1}{2}\le\delta<r_-$, we obtain*

$$1<\delta+r_{ij}\le r_{ji}+r_{ij}=r_i+r_j \ \text{ (for all } i,j),$$

and

$$\left\langle|\lambda|^{1/r^+}\xi,|\lambda|^{1/r_i}\right\rangle^{-r_i} \le \frac{1}{\big(|\lambda|^{2/r^+}|\xi|^2+|\lambda|^{2/r_i}\big)^{r_i/2}}$$
$$\le \frac{1}{\big(|\lambda|^{2/r^+}|\xi|^2+|\lambda|^{2/r^+}\big)^{r_i/2}}$$
$$= |\lambda|^{-r_i/r^+}\langle\xi\rangle^{-r_i} \tag{11}$$

for all $\xi \in \mathbb{R}^n$ *and* $\lambda \in \Lambda_\omega$. *Moreover,* $\sigma := \frac{r_-}{r^+} \in (1/2r^+, 1]$ *and* $\mu := |\lambda|^{1/r^+}$, *with* $\lambda \in \Lambda_\omega$, *satisfies*

$$\mu^{-r_{ij}} = \frac{1}{|\lambda|^{r_{ij}/r^+}} \leq \frac{1}{|\lambda|^\sigma} \quad \text{for all } i, j.$$

Now, we will establish a key lemma for the generation of analytic semigroup. In the lemma, σ and μ are as in Remark 3.

Lemma 4. *Let* $\frac{1}{2} \leq \delta < \min\{1, r_-\}$ *and* K_ε *as in* (9). *Then:*

(a) *There exists a constant* $C > 0$ *such that for all* $\varepsilon \in (0,1)$, $\eta \in \mathbb{R}^n$ *and* $\lambda \in \Lambda_\omega$ *it holds*

$$(1 + |\mu\eta|)|\mu\eta|^n \|K_\varepsilon(\eta, \lambda)\|_{\mathcal{L}(\mathbb{C}^q)} \leq \frac{C}{|\lambda|^\sigma} \mu^n |\mu\eta|^\delta. \tag{12}$$

(b) *There exists a strongly measurable function* $K : \mathbb{R}^n \times \Lambda_\omega \to \mathcal{L}(\mathbb{C}^q)$ *with* $K_\varepsilon(\eta, \lambda) \to K(\eta, \lambda)$ $(\varepsilon \searrow 0)$ *pointwise, and the estimate* (12) *holds with* K_ε *being replaced by* K. *In consequence there exists a constant* $M > 0$, *independent on* λ, *such that*

$$\|K(\cdot, \lambda)\|_{L^1(\mathbb{R}^n, \mathcal{L}(\mathbb{C}^q))} \leq \frac{M}{|\lambda|^\sigma} \quad \forall \lambda \in \Lambda_\omega. \tag{13}$$

Proof. (a) First, note that with the change $\xi \mapsto \mu\xi$ we obtain

$$K_\varepsilon(\eta, \lambda) = \mu^n \int_{\mathbb{R}^n} e^{i\mu\xi\cdot\eta} \chi_\varepsilon(\mu\xi, \eta; \lambda) b_\lambda(\mu\xi) \frac{d\xi}{(2\pi)^n}.$$

Note also that, for $\alpha \in \mathbb{N}_0^n$ with $0 < |\alpha| \leq \rho$, it holds

$$\int_{\mathbb{R}^n} D_\xi^\alpha (\chi_\varepsilon(\mu\xi, \eta; \lambda) b_\lambda(\mu\xi)) \frac{d\xi}{(2\pi)^n} = 0.$$

With this, $\left|e^{i\mu\xi\cdot\eta} - 1\right| \leq 2|\mu\xi|^\delta |\eta|^\delta$ for all $\xi, \eta \in \mathbb{R}^n$ and $\delta \in (0,1)$, partial integration, Leibniz rule, (10), corollary 1, Lemma 3, and Remark 3, we obtain for all $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = n + l$, $l = 0, 1$, and $\frac{1}{2} \leq \delta < \min\{1, r_-\}$, that

$$
\begin{aligned}
&\left\|(\mu\eta)^{\alpha}K_{\varepsilon}(\eta,\lambda)\right\|_{\mathcal{L}(\mathbb{C}^q)}\\
&=\left\|\mu^n\int_{\mathbb{R}^n}e^{i\mu\xi\cdot\eta}D_{\xi}^{\alpha}(\chi_{\varepsilon}(\mu\xi,\eta;\lambda)b_{\lambda}(\mu\xi))\frac{d\xi}{(2\pi)^n}\right\|_{\mathcal{L}(\mathbb{C}^q)}\\
&=\left\|\mu^n\int_{\mathbb{R}^n}(e^{i\mu\xi\cdot\eta}-1)\psi(\varepsilon\eta)D_{\xi}^{\alpha}(\chi_{\varepsilon}(\mu\xi;\lambda)b_{\lambda}(\mu\xi))\frac{d\xi}{(2\pi)^n}\right\|_{\mathcal{L}(\mathbb{C}^q)}\\
&\le\mu^n\int_{\mathbb{R}^n}2|\mu\xi|^{\delta}|\eta|^{\delta}|\psi(\varepsilon\eta)|\sum_{\gamma\le\alpha}C_{\gamma\alpha}\left|D_{\xi}^{\alpha-\gamma}(\chi_{\varepsilon}(\mu\xi;\lambda))\right|\left\|\partial_{\xi}^{\gamma}(b_{\lambda}(\mu\xi))\right\|_{\mathcal{L}(\mathbb{C}^q)}d\xi\\
&\le 2\mu^n|\mu\eta|^{\delta}\int_{\mathbb{R}^n}|\xi|^{\delta}\sum_{\gamma\le\alpha}C_{\gamma\alpha}\mu^{|\alpha|}C_{\alpha-\gamma}\langle\mu\xi,\mu\rangle^{|\gamma|-|\alpha|}\left\|(\partial_{\xi}^{\gamma}b_{\lambda})(\mu\xi)\right\|_{\mathcal{L}(\mathbb{C}^q)}d\xi\\
&\le 2\mu^n|\mu\eta|^{\delta}\int_{\mathbb{R}^n}|\xi|^{\delta}\sum_{i,j}\sum_{\gamma\le\alpha}\overline{C}_{\gamma\alpha}\mu^{|\alpha|}C_{\alpha-\gamma}\langle\mu\xi,\mu\rangle^{|\gamma|-|\alpha|}\left|(\partial_{\xi}^{\gamma}g_{ij}(\xi;\lambda))(\mu\xi)\right|d\xi\\
&\le C\mu^n|\mu\eta|^{\delta}\int_{\mathbb{R}^n}|\xi|^{\delta}\Big[\sum_{i=1}^{q}\widehat{C}_{0\alpha}\mu^{|\alpha|}\langle\mu\xi,\mu\rangle^{-|\alpha|}\left\langle\mu\xi,|\lambda|^{1/r_i}\right\rangle^{-r_i}\\
&\quad+\sum_{j=1}^{q}\sum_{\mathcal{Z}_1}\widehat{C}_{\alpha}\mu^{|\alpha|}\langle\mu\xi,\mu\rangle^{|\gamma|-|\alpha|}\langle\mu\xi\rangle^{r_j-|\gamma|}\left\langle\mu\xi,|\lambda|^{1/r_i}\right\rangle^{-r_i}\left\langle\mu\xi,|\lambda|^{1/r_j}\right\rangle^{-r_j}\\
&\quad+\sum_{\mathcal{Z}_2}\widehat{C}_{\gamma\alpha}\mu^{|\alpha|}\langle\mu\xi,\mu\rangle^{|\gamma|-|\alpha|}\langle\mu\xi\rangle^{r_{ij}-|\gamma|}\left\langle\mu\xi,|\lambda|^{1/r_i}\right\rangle^{-r_i}\left\langle\mu\xi,|\lambda|^{1/r_j}\right\rangle^{-r_j}\Big]d\xi\\
&\overset{(11)}{\le} C\mu^n|\mu\eta|^{\delta}\int_{\mathbb{R}^n}|\xi|^{\delta}\Big[\sum_{i=1}^{q}\widehat{C}_{0\alpha}\mu^{|\alpha|}|\mu\xi,\mu|^{-|\alpha|}\mu^{-r_i}\langle\xi\rangle^{-r_i}\\
&\qquad+\sum_{j=1}^{q}\sum_{\mathcal{Z}_1}\widehat{C}_{\alpha}\mu^{|\alpha|}\langle\mu\xi\rangle^{r_j-|\alpha|}\mu^{-r_i}\langle\xi\rangle^{-r_i}\mu^{-r_j}\langle\xi\rangle^{-r_j}\\
&\qquad+\sum_{\mathcal{Z}_2}\widehat{C}_{\gamma\alpha}\mu^{|\alpha|}\langle\mu\xi\rangle^{r_{ij}-|\alpha|}\mu^{-r_i}\langle\xi\rangle^{-r_i}\mu^{-r_j}\langle\xi\rangle^{-r_j}\Big]d\xi\\
&\le\overline{C}\mu^n|\mu\eta|^{\delta}\Big[\frac{1}{|\lambda|^{\sigma}}\sum_{i=1}^{q}\underbrace{\int_{\mathbb{R}^n}\langle\xi\rangle^{-(r_i+n+l-\delta)}d\xi}_{=:C_{il}<\infty}\\
&\qquad+\sum_{i,j=1}^{q}\underbrace{\int_{\mathbb{R}^n}\mu^{n+l-r_i-r_j}|\xi|^{\delta}\langle\mu\xi\rangle^{r_{ij}-(n+l)}\langle\xi\rangle^{-r_i-r_j}d\xi}_{=:I_{ijl}}\Big].
\end{aligned}
$$

Let $\Omega_1:=\{\xi\in\mathbb{R}^n:|\xi|<1\}$, $\Omega_2:=\mathbb{R}^n\backslash\Omega_1$ and

$$I_{ijl}^{(k)}:=\int_{\Omega_k}\mu^{n+l-r_i-r_j}|\xi|^{\delta}\langle\mu\xi\rangle^{r_{ij}-(n+l)}\langle\xi\rangle^{-r_i-r_j}d\xi$$

for $k=1,2$. Since $I_{ijl}=I_{ijl}^{(1)}+I_{ijl}^{(2)}$, we will estimate $I_{ijl}^{(k)}$. We consider two cases: Case 1. If $r_{ij}\le n+l$ for some i,j, it holds

$$I_{ijl}^{(1)} \leq \int_{\Omega_1} \mu^{n+l-r_i-r_j}|\xi|^{\delta}\mu^{r_{ij}-(n+l)}|\xi|^{r_{ij}-(n+l)}d\xi$$
$$= \mu^{-r_{ji}}\int_{\Omega_1}|\xi|^{\delta+r_{ij}-(n+l)}d\xi \leq \frac{C}{|\lambda|^{\sigma}},$$

since $\frac{1}{2} \leq \delta < r_{ij}$ (thus $\delta + r_{ij} > 1$). Furthermore,

$$I_{ijl}^{(2)} \leq \int_{\Omega_2} \mu^{n+l-r_i-r_j}|\xi|^{\delta}\mu^{r_{ij}-(n+l)}|\xi|^{r_{ij}-(n+l)}|\xi|^{-r_i-r_j}d\xi$$
$$= \mu^{-r_{ji}}\int_{\Omega_2}|\xi|^{\delta-r_{ji}-n-l}d\xi \leq \frac{C}{|\lambda|^{\sigma}},$$

due to $\delta < r_{ji} + l$ for $l = 0, 1$. Therefore,

$$I_{ijl} \leq \frac{\widehat{C}}{|\lambda|^{\sigma}} \qquad (l = 0, 1). \tag{14}$$

Case 2. Suppose $r_{ij} > n + l$ for some i, j. Since $\mu \geq 1$, then we get

$$I_{ijl}^{(1)} \leq \int_{\Omega_1} \mu^{n+l-r_i-r_j}|\xi|^{\delta}\left(1+\mu^2|\xi|^2\right)^{\frac{r_{ij}-(n+l)}{2}}d\xi$$
$$\leq \int_{\Omega_1} \mu^{n+l-r_i-r_j}|\xi|^{\delta}2^{\frac{r_{ij}-n-l}{2}}\mu^{r_{ij}-n-l}d\xi$$
$$= 2^{\frac{r_{ij}-n-l}{2}}\mu^{-r_{ji}}\int_{\Omega_1}|\xi|^{\delta}d\xi \leq \frac{C}{|\lambda|^{\sigma}}.$$

Moreover,

$$I_{ijl}^{(2)} = \int_{\Omega_2} \mu^{n+l-r_i-r_j}|\xi|^{\delta}\left(1+\mu^2|\xi|^2\right)^{\frac{r_{ij}-(n+l)}{2}}\langle\xi\rangle^{-r_i-r_j}d\xi$$
$$\leq \int_{\Omega_2} \mu^{n+l-r_i-r_j}|\xi|^{\delta}|\xi|^{r_{ij}-(n+l)}\left(1+\mu^2\right)^{\frac{r_{ij}-(n+l)}{2}}|\xi|^{-r_i-r_j}d\xi$$
$$\leq \int_{\Omega_2} \mu^{n+l-r_i-r_j}|\xi|^{\delta-r_{ji}-(n+l)}2^{\frac{r_{ij}-n-l}{2}}\mu^{r_{ij}-n-l}d\xi$$
$$= 2^{\frac{r_{ij}-n-l}{2}}\mu^{-r_{ji}}\int_{\Omega_2}|\xi|^{\delta-r_{ji}-n-l}d\xi \leq \frac{C}{|\lambda|^{\sigma}}.$$

Thus, (14) holds too. In consequence

$$\left\|(\mu\eta)^{\alpha}K_{\varepsilon}(\eta,\lambda)\right\|_{\mathcal{L}(\mathbb{C}^q)} \leq \frac{C}{|\lambda|^{\sigma}}\mu^n|\mu\eta|^{\delta}$$

for all $\eta \in \mathbb{R}^n$, $\lambda \in \Lambda_\omega$, $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = n + l$, $l = 0, 1$, $\varepsilon \in (0, 1)$ and $\frac{1}{2} \leq \delta < \min\{1, r_-\}$. Therefore, we have

$$|\mu\eta|^{n+l} \|K_\varepsilon(\eta, \lambda)\|_{\mathcal{L}(\mathbb{C}^q)} \leq n^{\frac{n+l}{2}} \sum_{|\alpha|=n+l} \left\|(\mu\eta)^\alpha K_\varepsilon(\eta, \lambda)\right\|_{\mathcal{L}(\mathbb{C}^q)} \leq \frac{C}{|\lambda|^\sigma} \mu^n |\mu\eta|^\delta.$$

Adding these inequalities for $l = 0$ and $l = 1$, we obtain the assertion (a).
(b) Let $\varepsilon, \varepsilon' \in (0, 1)$, $\eta \in \mathbb{R}^n$ and $\lambda \in \Lambda_\omega$. From the proof of (a) we see that

$$\begin{aligned} &(\mu\eta)^\alpha (K_\varepsilon(\eta, \lambda) - K_{\varepsilon'}(\eta, \lambda)) \\ &= \mu^n \int_{\mathbb{R}^n} (e^{i\mu\xi\cdot\eta} - 1) D_\xi^\alpha \left[(\chi_\varepsilon(\mu\xi, \eta; \lambda) - \chi_{\varepsilon'}(\mu\xi, \eta; \lambda)) b_\lambda(\mu\xi) \right] \frac{d\xi}{(2\pi)^n}. \end{aligned} \tag{15}$$

From Lemma 3 we know that $D_\xi^\gamma(\chi_\varepsilon(\mu\xi, \eta; \lambda) - \chi_{\varepsilon'}(\mu\xi, \eta; \lambda)) \longrightarrow 0$ $(\varepsilon, \varepsilon' \searrow 0)$ for all $\gamma \in \mathbb{N}_0^n$ and all ξ, η. Therefore the integrand in (15) converges pointwise to zero for $\varepsilon, \varepsilon' \searrow 0$. Furthermore, in the same way of the proof of part (a) we have that

$$\begin{aligned} &\left\| (e^{i\mu\xi\cdot\eta} - 1) D_\xi^\alpha [(\chi_\varepsilon(\mu\xi, \eta; \lambda) - \chi_{\varepsilon'}(\mu\xi, \eta; \lambda)) b_\lambda(\mu\xi)] \right\|_{\mathcal{L}(\mathbb{C}^q)} \\ &\leq C|\mu\eta|^\delta \Big[\frac{1}{|\lambda|^\sigma} \sum_{i=1}^q \langle\xi\rangle^{-(r_i+n+l-\delta)} \\ &\qquad + \sum_{i,j=1}^q \mu^{n+l-r_i-r_j} |\xi|^\delta \langle\mu\xi\rangle^{r_{ij}-(n+l)} \langle\xi\rangle^{-r_i-r_j} \Big] \in L^1\left(\mathbb{R}_\xi^n\right). \end{aligned}$$

Hence, by dominated convergence we get for fixed $(\eta, \lambda) \in (\mathbb{R}^n \setminus \{0\}) \times \Lambda_\omega$ that $\|K_\varepsilon(\eta, \lambda) - K_{\varepsilon'}(\eta, \lambda)\|_{\mathcal{L}(\mathbb{C}^q)} \longrightarrow 0$ $(\varepsilon, \varepsilon' \searrow 0)$. Therefore there exists a strongly measurable function $K : \mathbb{R}^n \times \Lambda_\omega \to \mathcal{L}(\mathbb{C}^q)$ with $K_\varepsilon(\eta, \lambda) \to K(\eta, \lambda)$ $(\varepsilon \searrow 0)$ pointwise a.e. Then, inequality (12) holds for $K(\eta, \lambda)$ instead of $K_\varepsilon(\eta, \lambda)$ and in consequence (13) is true due to

$$\int_{\mathbb{R}^n} \frac{\mu^n |\mu\eta|^{\delta-n}}{1 + |\mu\eta|} d\eta < \infty.$$

□

Proposition 1. *Let $A(D)$ be Λ-elliptic with $\rho \geq \rho_n$, $\frac{1}{2} < r_-$ and let $b_\lambda(\cdot) := (A_{\alpha_0}(\cdot) - \lambda)^{-1}$ for all $\lambda \in \Lambda_\omega$. If $k \in \mathbb{N}_0$ and $1 \leq p < \infty$, then $b_\lambda(D) \in \mathcal{L}\left(W_p^k(\mathbb{R}^n, \mathbb{C}^q)\right)$ with*

$$\|b_\lambda(D)\|_{\mathcal{L}\left(W_p^k(\mathbb{R}^n, \mathbb{C}^q)\right)} \leq \frac{M}{|\lambda|^\sigma} \quad \forall \lambda \in \Lambda_\omega,$$

where the constant $M > 0$ is independent on λ and σ.

Proof. Let $\beta \in \mathbb{N}_0^n$ with $|\beta| \leq k$, $\lambda \in \Lambda_\omega$, $u \in C_b^\infty(\mathbb{R}^n, \mathbb{C}^q) \cap W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ and $x \in \mathbb{R}^n$. Then (see (8))

$$\partial_x^\beta (b_\lambda(D)u)(x) = \lim_{\varepsilon \searrow 0} \int_{\mathbb{R}^n} K_\varepsilon(\eta, \lambda)(\partial_x^\beta u)(x - \eta) d\eta \tag{16}$$

with K_ε as in (9). From (16), Lemma 4 and dominated convergence, we get

$$\partial_x^\beta (b_\lambda(D)u)(x) = \int_{\mathbb{R}^n} K(\eta, \lambda)(\partial_x^\beta u)(x - \eta) d\eta = (K(\cdot, \lambda) * (\partial_x^\beta u))(x),$$

where $*$ stands for the standard convolution. Since $\partial_x^\beta u \in L^p(\mathbb{R}^n, \mathbb{C}^q)$, we have $K(\cdot, \lambda) * (\partial_x^\beta u) \in L^1(\mathbb{R}^n, \mathbb{C}^q)$ and

$$\begin{aligned}\left\|\partial_x^\beta (b_\lambda(D)u)\right\|_{L^p(\mathbb{R}^n,\mathbb{C}^q)} &\leq \|K(\cdot,\lambda)\|_{L^1(\mathbb{R}^n,L(\mathbb{C}^q))}\|\partial_x^\beta u\|_{L^p(\mathbb{R}^n,\mathbb{C}^q)} \\ &\leq \frac{M}{|\lambda|^\sigma}\|u\|_{W_p^k(\mathbb{R}^n,\mathbb{C}^q)}\end{aligned}$$

due to Lemma 4 (b). This implies that

$$\|b_\lambda(D)u\|_{W_p^k(\mathbb{R}^n,\mathbb{C}^q)} \leq \frac{\widetilde{M}}{|\lambda|^\sigma}\|u\|_{W_p^k(\mathbb{R}^n,\mathbb{C}^q)} \tag{17}$$

for all $u \in C_b^\infty(\mathbb{R}^n, \mathbb{C}^q) \cap W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ and $\lambda \in \Lambda_\omega$. Because of $1 \leq p < \infty$, $\mathcal{S}(\mathbb{R}^n, \mathbb{C}^q)$ is dense in $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ which gives $b_\lambda(D) \in \mathcal{L}\left(W_p^k(\mathbb{R}^n, \mathbb{C}^q)\right)$ and the estimate on its norm. □

For $k \in \mathbb{N}_0$ and $1 \leq p < \infty$, we define the $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$-realization $A_{\alpha_0,k}$ of the system $A_{\alpha_0}(D)$ as the unbounded operator given by

$$\begin{aligned}D(A_{\alpha_0,k}) &:= \left\{u \in W_p^k(\mathbb{R}^n,\mathbb{C}^q) : A_{\alpha_0}(D)u \in W_p^k(\mathbb{R}^n,\mathbb{C}^q)\right\}, \\ A_{\alpha_0,k}u &:= A_{\alpha_0}(D)u \text{ for } u \in D(A_{\alpha_0,k}).\end{aligned}$$

Now we are able to show the main result of this paper. We recall that $\rho \geq \rho_n$, $\frac{1}{2} < r_-$ and $\sigma = \frac{r_-}{r^+} \in (\frac{1}{2r^+}, 1]$.

Theorem 1. *Let $A(D)$ be $\Lambda(\theta)$-elliptic with $0 < \theta < \pi/2$ and $\vartheta := \pi - \theta$. Let $k \in \mathbb{N}_0$, $1 \leq p < \infty$ and $A_{\alpha_0,k}$ be the $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$-realization of $A_{\alpha_0}(D)$. Then, for the resolvent set $\rho(-A_{\alpha_0,k})$ of $-A_{\alpha_0,k}$ we have $\rho(-A_{\alpha_0,k}) \supset \Sigma_{\vartheta,\omega} := \{\lambda \in \mathbb{C} : |\lambda| \geq \omega \text{ and } |\arg\lambda| \leq \vartheta\}$ and*

$$\|(\lambda + A_{\alpha_0,k})^{-1}\|_{\mathcal{L}(W_p^k(\mathbb{R}^n,\mathbb{C}^q))} \leq \frac{M}{|\lambda|^\sigma} \qquad (\lambda \in \Sigma_{\vartheta,\omega}). \tag{18}$$

for some constant $M > 0$. Therefore, $-A_{\alpha_0,k} : W_p^k(\mathbb{R}^n, \mathbb{C}^q) \supset D(A_{\alpha_0,k}) \to W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ generates an infinitely differentiable semigroup on $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$, which is analytic and strongly continuous if $\sigma = 1$ (i.e., $r_1 = \cdots = r_q = r_-$).

Remark 4. *The semigroup is given by $(e^{-\tau A_{\alpha_0,k}})_{\tau \geq 0}$ with $e^{-0A_{\alpha_0,k}} := I$ and*

$$e^{-\tau A_{\alpha_0,k}} := \frac{1}{2\pi i}\int_\Gamma e^{-\tau\lambda}(\lambda I - A_{\alpha_0,k})^{-1}d\lambda \quad (\tau > 0),$$

where $\Gamma : \lambda = \omega + iy$, $-\infty < y < \infty$ stands for a lying in $\rho(-A_{\alpha_0,k})$ path, and $[t \mapsto e^{-\tau A_{\alpha_0,k}}] \in C^\infty((0,\infty); \mathcal{L}(W_p^k(\mathbb{R}^n, \mathbb{C}^q)))$. See [21, Theorem 3.4, Ch. 1] for a reference. Further results about differential and analytical properties of semigroups of operators can be found also in [22] and in the references therein.

Proof of Theorem 1. Because of the density of $\mathcal{S}(\mathbb{R}^n, \mathbb{C}^q)$ in $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ and

$$(A_{\alpha_0,k} - \widetilde{\lambda})b_{\widetilde{\lambda}}(D)u = b_{\widetilde{\lambda}}(D)(A_{\alpha_0,k} - \widetilde{\lambda})u = u$$

for all $u \in \mathcal{S}(\mathbb{R}^n, \mathbb{C}^q)$ and $\widetilde{\lambda} \in \Lambda(\theta)_\omega$, it follows from (17) that $\Lambda(\theta)_\omega \subset \rho(A_{\alpha_0,k})$ and $b_{\widetilde{\lambda}}(D) = (A_{\alpha_0,k} - \widetilde{\lambda})^{-1}$ in $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$. Now, if $\lambda \in \Sigma_{\vartheta,\omega}$, then $\widetilde{\lambda} := -\lambda \in \Lambda(\theta)_\omega$. Therefore we have

$$(\lambda + A_{\alpha_0,k})b_{-\lambda}(D)u = b_{-\lambda}(D)(\lambda + A_{\alpha_0,k})u = u$$

for all $u \in \mathcal{S}(\mathbb{R}^n, \mathbb{C}^q)$ and $\lambda \in \Sigma_{\vartheta,\omega}$. It follows that $\Sigma_{\vartheta,\omega} \subset \rho\left(-A_{\alpha_0,k}\right)$ and $b_{-\lambda}(D) = \left(\lambda + A_{\alpha_0,k}\right)^{-1}$ for $\lambda \in \Sigma_{\vartheta,\omega}$. Then (18) follows from Proposition 1. □

The above result on the generation of semigroup in $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ allow us to solve non-autonomous Cauchy problems, based on an abstract result in [23], Chapter IV. For this, let $T > 0$ and assume $\mathcal{A} = \{A(t, D) : t \in [0, T]\}$ to be a uniformly bounded family of Λ-elliptic systems. For $k \in \mathbb{N}_0$ and $1 \leq p < \infty$, we denote by $A_k(t)$ the $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$-realization of $A(t, D)$. Then, we study the Cauchy problem

$$\begin{cases} \partial_t u(t) + A_k(t)u(t) = f(t), & t \in (0, T], \\ u(0) = u_0. \end{cases} \tag{19}$$

A function $u \in C^1\Big((0, T], W_p^k(\mathbb{R}^n, \mathbb{C}^q)\Big) \cap C\Big([0, T], W_p^k(\mathbb{R}^n, \mathbb{C}^q)\Big)$ is called a classical solution of (19), if $u(t) \in D(A_k(t))$ for all $t \in (0, T]$, $\partial_t u(t) + A_k(t)u(t) = f(t)$ for all $t \in (0, T]$ and $u(0) = u_0$.

Using Theorem 1 and the abstract result on Cauchy problems given in Theorem 2.5.1 of Chapter IV in [23], we obtain, in the same way to the proof of Theorem 4.3 in [6], the following result.

Theorem 2. *Let $\mathcal{A} = \{A(t, D) : t \in [0, T]\}$ be a uniformly bounded family of $\Lambda(\theta)$-elliptic systems, $0 < \theta < \pi/2$, with symbols $\left(a_{ij}(t, \xi)\right)_{1 \leq i,j \leq q}$ for all $t \in [0, T]$, such that $\left[t \mapsto a_{ij}(t, \cdot)\right] \in C^{\alpha}([0, T], S^{r_{ij},\rho}(\mathbb{R}^n))$ for all $i, j = 1, ..., q$ and some $\alpha \in (0, 1)$, with $r_1 = \cdots = r_q = r_- > 1/2$. Furthermore, suppose that there exists $\alpha_0 \in \mathbb{R}$ such that $A_{\alpha_0}(t, D) := A(t, D) + \alpha_0$ is $\Lambda(\theta)$-elliptic, $0 < \theta < \pi/2$, with the same constant C and $R = 0$, for all $t \in [0, T]$ (see Definition 3 and Remark 2). Moreover, let $k \in \mathbb{N}_0$, $1 \leq p < \infty$ and $\varepsilon \in (0, 1)$. Then, for every $u_0 \in W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ and $f \in C^{\varepsilon}\Big([0, T], W_p^k(\mathbb{R}^n, \mathbb{C}^q)\Big)$, the Cauchy problem*

$$\begin{cases} \partial_t v(t) + A_{\alpha_0,k}(t)v(t) = e^{-\alpha_0 t} f(t), & t \in (0, T], \\ v(0) = u_0. \end{cases} \tag{20}$$

has a unique classical solution, where $A_{\alpha_0,k}(t)$ is the $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$-realization of $A_{\alpha_0}(t, D)$.

Corollary 2. *Suppose that the same hypothesis from Theorem 2 hold. Then, there exists a unique classical solution of problem (19).*

Proof. First note that $A_{\alpha_0,k}(t) = A_k(t) + \alpha_0$. Now, let $v(t)$, $t \in [0, T]$, be the classical solution of problem (20) and set $u(t) := e^{\alpha_0 t} v(t)$ for $t \in [0, T]$. Then u is the unique classical solution of problem (19). □

Remark 5. *If $-A_{\alpha_0,k}(t)$, $t \in [0, T]$, generates only an infinitely differentible semigroup on $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ and, $A_{\alpha_0,k}(\cdot)^{-1}$ is strongly continuously differentiable on $[0, T]$ and satisfies some additional conditions, Theorems 4.3, 4.4, and Remark 4.5 in [24] imply the existence and uniqueness of a strict solution of (20), and therefore of (19), for each $u_0 \in W_p^k(\mathbb{R}^n, \mathbb{C}^q)$. Such strict solution is taken in sense of Definition 1.1 in [24].*

Remark 6. *(i) With the method used in this paper some better assertions could be obtained, for instance maximal L^p-regularity or the existence of a H_∞-calculus as in [4].*

(ii) Using some ideas from [4], one could change the basic space $W_p^k(\mathbb{R}^n, \mathbb{C}^q)$ by $\prod_{i=1}^{q} W_p^{k-l_i}(\mathbb{R}^n)$ for some suitable integers l_i, $i = 1, \ldots, q$. Thus one could obtain similar result as in Theorem 1, but under weaker assumption on the structure of the system. This remark will be useful for the analysis, in a forthcoming paper, of the generalized thermoelastic plate equations with fractional damping.

5. Examples

In this section, we will consider some examples where we could apply our results. Initially, as a naive example, we consider the Cauchy problem associated to the n-dimensional linear heat equation in the whole space. That is

$$\begin{cases} u_t(x,t) - \alpha\Delta u(x,t) = 0 & (x \in \mathbb{R}^n, t > 0), \\ u(x,0) = u_0(x) & (x \in \mathbb{R}^n), \end{cases} \tag{21}$$

where $\alpha > 0$ is related to the thermal diffusivity and $u(x,t)$ represents the temperature in point x at time t. The differential equation in (21) can be written in the form

$$u_t - A(D)u = 0,$$

where

$$A(\xi) = -\alpha|\xi|^2, \quad \xi \in \mathbb{R}^n.$$

Note that in this case $r_1 = 2 > 1/2$ and therefore the condition (2) holds trivially. Let define

$$A_-(\xi) := -A(\xi).$$

Now, for all $\lambda \in \Lambda(\theta)$ with $0 < \theta < \pi/2$, and all $|\xi| \geq \frac{1}{\sqrt{3}}$, it can be shown that

$$|\det(\lambda - A_-(\xi))| = \left|\lambda - \alpha|\xi|^2\right| \geq M\big(\langle\xi\rangle^2 + |\lambda|\big).$$

Then $A(D)$ is $\Lambda(\theta)$-elliptic and we can apply corollary 2 to solve problem (21).

Consider now the thermoelastic plate equations on $\mathbb{R}^n$ given by

$$\begin{cases} v_{tt} + \Delta^2 v + \Delta\theta = 0, \\ \theta_t - \Delta\theta - \Delta v_t = 0 \end{cases} \tag{22}$$

together with the initial conditions

$$v(0,\cdot) = v_0, \ v_t(0,\cdot) = v_1, \ \theta(0,\cdot) = \theta_0.$$

The equations in (22) were derived in [25], where v denotes a mechanical variable representing the vertical displacement of the plate, while θ denotes a thermal variable describing the temperature relative to a constant reference temperature $\overline{\theta}$.

Using the substitution $u = (\theta, v_t, -\Delta v)^\top$, the system (22) can be written as

$$u_t - A(D)u = 0, \tag{23}$$

where

$$A(\xi) := \begin{pmatrix} -|\xi|^2 & -|\xi|^2 & 0 \\ |\xi|^2 & 0 & -|\xi|^2 \\ 0 & |\xi|^2 & 0 \end{pmatrix}. \tag{24}$$

Note that in this case, $r_{ij} = 2 > 1/2$ for all $i, j = 1, 2, 3$, and assumption (2) holds. Now, we define

$$A_-(\xi) := -A(\xi) \tag{25}$$

and consider the determinant of $\lambda - A_-(\xi)$, which is given by:

$$\det(\lambda - A_-(\xi)) = \lambda^3 - |\xi|^2\lambda^2 + 2|\xi|^4\lambda - |\xi|^6.$$

It is easy to see that

$$\det(\lambda - A_-(\xi)) = |\xi|^6 p\left(\frac{\lambda}{|\xi|^2}\right), \tag{26}$$

where

$$p(t) = t^3 - t^2 + 2t - 1.$$

Since $p(0) < 0,\ p(1) > 0$ and $p'(t) > 0$ for all $t \in \mathbb{R}$, there exists a unique real number $\alpha \in (0,1)$ such that $p(\alpha) = 0$. Now, since p is a polynomial with real coefficients, there exist positive constants β and γ, such that

$$p(t) = (t - \lambda_1)(t - \lambda_2)(t - \lambda_3) \tag{27}$$

with $\lambda_1 = \alpha$, $\lambda_2 = \beta + \gamma i$ and $\lambda_3 = \overline{\lambda_2}$. In particular, we get $\lambda_1 + \lambda_2 + \lambda_3 = 1$, and therefore $\beta = \frac{1-\alpha}{2} > 0$. Hence, according to (26) and (27), it follows that

$$\det(\lambda - A_-(\xi)) = \Big(\lambda - |\xi|^2\lambda_1\Big)\Big(\lambda - |\xi|^2\lambda_2\Big)\Big(\lambda - |\xi|^2\lambda_3\Big). \tag{28}$$

By inequality (2.7) in [26], there exists $\frac{\pi}{2} < \vartheta_0 < \pi$ such that

$$\left|\lambda\lambda_j^{-1} + |\xi|^2\right| \geq C(|\lambda| + |\xi|^2) \quad (j = 1,2,3) \quad \forall \lambda \in -\Lambda(\vartheta_0) \text{ and } \xi \in \mathbb{R}^n, \tag{29}$$

where $-\Lambda(\vartheta_0) := \{-\lambda : \lambda \in \Lambda(\vartheta_0)\}$. Hence, for all $\lambda \in \Lambda(\vartheta_0)$ and $|\xi| \geq \frac{1}{\sqrt{3}}$, we have

$$\begin{aligned}\left|\lambda - |\xi|^2\lambda_j\right| &= |\lambda_j|\left|(-\lambda)\lambda_j^{-1} + |\xi|^2\right| \\ &\geq c|\lambda_j|(|\lambda| + |\xi|^2) \\ &\geq C(|\lambda| + \langle\xi\rangle^2)\end{aligned} \tag{30}$$

for $j = 1,2,3$. Note that $2^r|\xi|^r \geq \langle\xi\rangle^r$ if $|\xi| \geq \frac{1}{\sqrt{3}}$ and $r \geq 0$.

Proposition 2. *Let $A_-(\xi)$ be defined as in (25). Then $A_-(D)$ is $\Lambda(\vartheta)$-elliptic with $0 < \vartheta < \pi/2$.*

Proof. This follows from (28)–(30). □

Theorem 3. *Let $T > 0$, $\varepsilon \in (0,1)$, $k \in \mathbb{N}_0$, $1 \leq p < \infty$, $A(D)$ be defined by (23) and (24) and let A_k be the $W_p^k(\mathbb{R}^n, \mathbb{C}^3)$-realization of $A(D)$. Then, for each $u_0 \in W_p^k(\mathbb{R}^n, \mathbb{C}^3)$ and $f \in C^\varepsilon\big([0,T], W_p^k(\mathbb{R}^n, \mathbb{C}^3)\big)$ the Cauchy problem*

$$\begin{cases} \partial_t u(t) + A_k u(t) = f(t), & t \in (0,T], \\ u(0) = u_0. \end{cases}$$

has a unique classical solution.

Proof. It follows from Proposition 2 and corollary 2. □

Now, as a third example, we consider the lineal structurally damped plate equation on $\mathbb{R}^n$

$$v_{tt} + \Delta^2 v - \rho\Delta v_t = f, \tag{31}$$

together with initial conditions

$$v(0,\cdot) = v_0,\ v_t(0,\cdot) = v_1.$$

Here, $\rho > 0$ is a fixed parameter. A description of this equation can be found in [27] and the references therein.

Using the substitution $u = (v_t, -\Delta v)^\top$ and $F = (f, 0)^T$, the Equation (31) can be written as

$$u_t - \mathscr{A}(D)u = F,$$

where

$$\mathscr{A}(\xi) := \begin{pmatrix} -\rho|\xi|^2 & -|\xi|^2 \\ |\xi|^2 & 0 \end{pmatrix}.$$

Note again that $r_{ij} = 2 > 1/2$ for all $i, j \in \{1, 2\}$, and assumption (2) holds. Now, we define $\mathscr{A}_-(\xi) := -\mathscr{A}(\xi)$ and consider the determinant of $\lambda - \mathscr{A}_-(\xi)$, which is given by

$$\det(\lambda - \mathscr{A}_-(\xi)) = \lambda^2 - \rho|\xi|^2\lambda + |\xi|^4.$$

Note that $\det(\lambda_\pm - \mathscr{A}_-(\xi)) = 0$ if only if $\lambda_\pm = |\xi|^2\left(\frac{\rho}{2} \pm \frac{\sqrt{\rho^2 - 4}}{2}\right) =: |\xi|^2 z_\pm$. If $\xi \neq 0$, then $\lambda_\pm > 0$ for $\rho \geq 2$ and $\lambda_\pm \in \mathbb{C}$ (with $\operatorname{Re} z_+ > 0$ and $z_- = \overline{z_+}$) for $0 < \rho < 2$. Therefore, $\det(\lambda - \mathscr{A}_-(\xi)) = \left(\lambda - |\xi|^2 z_+\right)\left(\lambda - |\xi|^2 z_-\right)$ and in consequence

$$|\det(\lambda - \mathscr{A}_-(\xi))| \geq C(\langle\xi\rangle^2 + |\lambda|)(\langle\xi\rangle^2 + |\lambda|) \ \ \forall \lambda \in \Lambda(\pi/2) \text{ and } |\xi| \geq 1/\sqrt{3}.$$

In consequence, $\mathscr{A}_-(D)$ is $\Lambda(\theta)$-elliptic with $0 < \theta < \pi/2$. Using the same arguments as in the previous example, we have that the Cauchy problem associated with (31) has a unique classical solution.

As a last example we consider a generalized plate equation in $\mathbb{R}^n$ with intermediated damping. Let $\alpha, \rho > 0$, $\beta \in [0, 1]$ and $L := (-\Delta)^\alpha$. Then the associated symbol of L is $p(\xi) = |\xi|^{2\alpha}$, $\xi \in \mathbb{R}^n$. The generalized plate equation in $\mathbb{R}^n$ with intermediated damping is given by

$$u_{tt} + Lu + \rho L^\beta u_t = 0 \tag{32}$$

together with the initial conditions

$$u(0, \cdot) = u_0, \ u_t(0, \cdot) = u_1. \tag{33}$$

The generalized thermoelastic plate equation has been introduced in [28], a plate equation with intermediate damping was studied in [29] and a plate equation with intermediate rotational force and damping in [30]. For the particular case $\alpha = 2$, (32) models the equation of a plate with: (i) frictional damping if $\beta = 0$, (ii) structural damping if $\beta = 1/2$ and (iii) Kelvin-Voigt damping if $\beta = 1$.

If $U := \left(u_t, L^{1/2}u\right)$, the equation (32) can equivalently be written as

$$U_t + \widetilde{A}(D)U = 0, \qquad \widetilde{A}(\xi) := \begin{bmatrix} \rho|\xi|^{2\alpha\beta} & |\xi|^\alpha \\ -|\xi|^\alpha & 0 \end{bmatrix}.$$

Now, let $\chi(\xi)$ be an arbitrary 0-excision function and $A(\xi) := \chi(\xi)\widetilde{A}(\xi)$. In the following we will omit without loss of generality the factor $\chi(\xi)$ in the definition of $A(\xi)$ and we will assume that $\rho = 1$.

Using the ideas of the proof of Lemma 6.1 in [4] we obtain the following lemma.

Lemma 5. *Assume that the parameters $\alpha > 0$ and $\beta \in (0, 1)$ satisfy the conditions*

$$\alpha > \frac{1}{2} \quad \wedge \quad \frac{1}{4\alpha} < \beta < 1 - \frac{1}{4\alpha}.$$

Then, for the the following choice of orders:

$$r_1 = 2\alpha\beta,\ r_2 = 2\alpha(1-\beta) \text{ and } r_{12} = r_{21} = \alpha,$$

$A(D)$ *is* $\Lambda(\theta)$*-elliptic for any* $0 < \theta < \pi$.

Under the hypotheses of the previous lemma we have that

$$r_+ = \begin{cases} 2\alpha(1-\beta), & 0 < \beta < \frac{1}{2}, \\ \alpha, & \beta = \frac{1}{2}, \\ 2\alpha\beta. & \frac{1}{2} < \beta < 1, \end{cases} \quad r_- = \begin{cases} 2\alpha\beta, & 0 < \beta < \frac{1}{2}, \\ \alpha, & \beta = \frac{1}{2}, \\ 2\alpha(1-\beta), & \frac{1}{2} < \beta < 1, \end{cases} \quad \text{and}$$

$$\sigma = \frac{r_-}{r_+} = \begin{cases} \frac{\beta}{1-\beta}, & 0 < \beta < \frac{1}{2}, \\ 1, & \beta = \frac{1}{2}, \\ \frac{1-\beta}{\beta}, & \frac{1}{2} < \beta < 1. \end{cases}$$

Consequently, we can apply corollary 2 and Remark 5 to solve problem (32) and (33).

6. Conclusions

In this article, we have proved that the additive inverse of a suitable Sobolev space realization of a Λ-elliptic Fourier multipliers system (in the sense of the Definition 3) generates an infinitely differentiable semigroup on such Sobolev space, and that under certain additional conditions, it generates an analytic semigroup on the same Sobolev space (see Theorem 1). We emphasize again in these conclusions that the proof of the generation of semigroups was done directly using an approach based on oscillatory integrals and non trivial kernel estimates for them. With the results about generation of semigroups we addressed the analysis of some application problems in Section 5 using well-known statements for the existence and uniqueness of solutions for abstract evolution equations. Now, regarding the possible future scope of this work, we recall Remark 6: using techniques similar to those in this paper, questions about maximal L^p-regularity, the existence of a H_∞-calculus, the improvement of the basic spaces, and the weakening of the assumptions for the structure of the system of Fourier multipliers, would be addressed in a forthcoming paper. In the other direction, it is interesting to study assumptions, under which Λ-elliptic Fourier multipliers systems generate Cosine families of operators in some appropriate functional or distributional spaces, to consider control problems for fractional evolution inclusions or equations following ideas from, for example [9,11–14].

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